



Mini-BEHAVIOR-Gran: Revealing U-Shaped Effects of Instruction Granularity on Language-Guided Embodied Agents

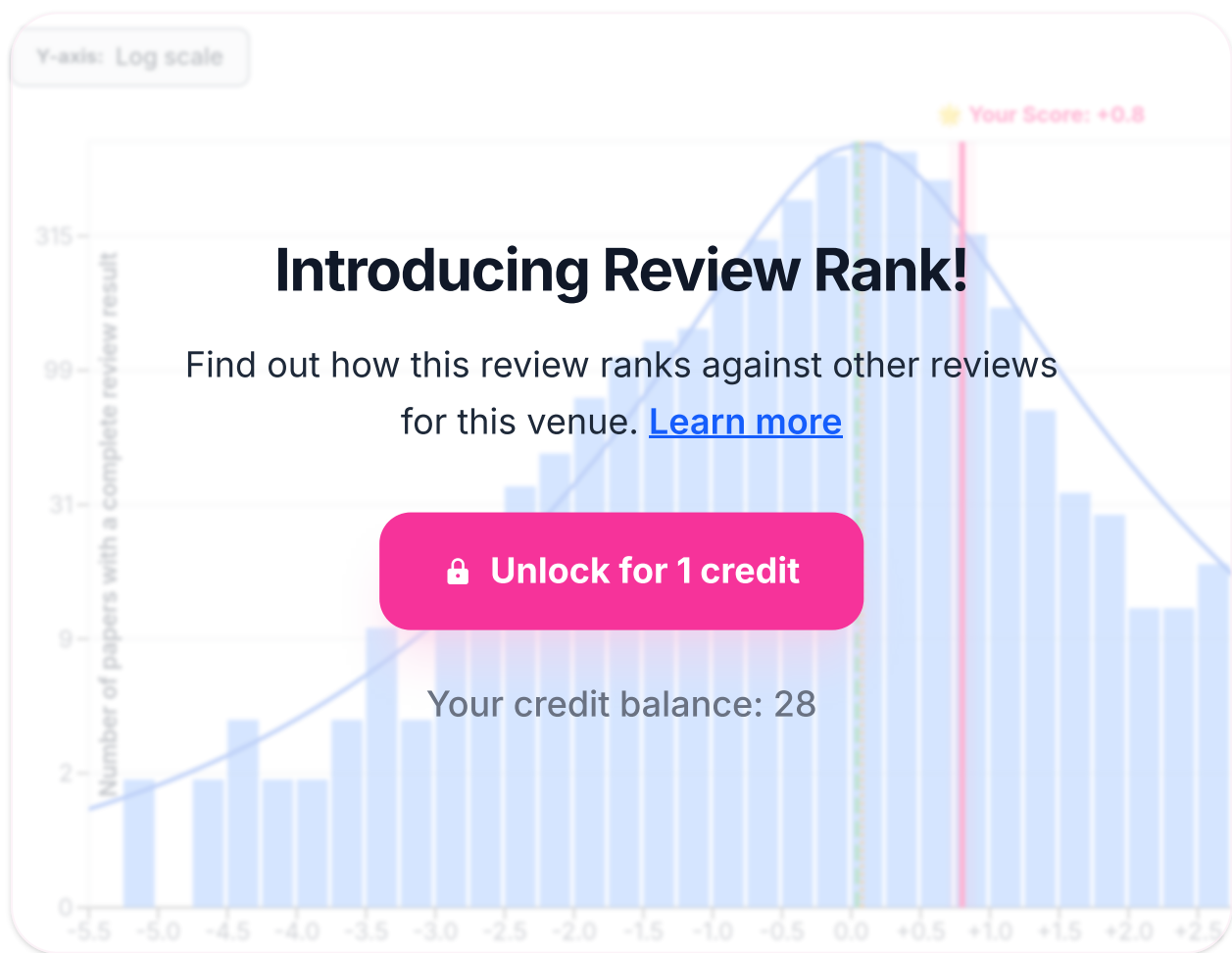
Anonymized Authors

Reference Check New

Copy View Paper



Reviewed as if submitted to CONFERENCE ACL 2026 - main track ⓘ



Desk Rejection Assessment:

Paper Length

Pass .

Topic Compatibility

Pass . The paper squarely targets multimodality and language grounding for embodied agents, with a strong NLP component in instruction representation, generation, and analysis. It aligns with ACL areas such as Multimodality and Language Grounding, Resources and Evaluation, and Interpretability/Analysis.

Minimum Quality

Pass . The submission includes Abstract, Introduction, Method/Benchmark construction (Sections 2–3), Experiments and Results (Section 4), Related Work (Section 5), Conclusion (Section 6), and a Limitations section (Section 7). The paper is technically sound overall, presents quantitative analyses, and is written in clear English.

Prompt Injection and Hidden Manipulation Detection

Pass . I found no signs of prompt injection or hidden instructions aimed at manipulating the review.

Expected Review Outcome:

Paper Summary

The paper introduces Mini-BEHAVIOR-Gran, a benchmark designed to study instruction granularity in language-guided embodied agents. For each of 20 long-horizon tasks, it provides multiple instruction variants per task and a dynamic instructor that issues rule-conditioned guidance as the agent progresses. The authors evaluate four candidate cross-task granularity metrics (token count, entity count, action-verb count, and planning width) and find width to be a more consistent proxy for difficulty. Using width-based grouping, they report a U-shaped performance curve where agents perform best with very fine or very coarse instructions, and worst under medium granularity. Analyses of instruction predictability and attention trends suggest the coarse-end rebound is associated with shallow grounding and vision-dominant policies.

Summary Of Strengths

- Clear benchmark design that isolates instruction granularity while holding task semantics constant. Section 2 details a shared symbolic feature space and a

≡ rule-based instruction representation that supports dynamic instruction updates.

- The planning-inspired “width” metric is well motivated as a cross-task proxy for latent coordination burden, and its computation is explained with a worked example. Table 3 and its accompanying explanation concretely illustrate how width scales with concurrent state features (e.g., cleaning k shoes gives $w = k+2$).
- Empirical evidence supports the central claims:
 - Figure 4 shows that width correlates negatively and consistently with success rate across instruction-horizon bins, while surface metrics behave inconsistently. This directly backs the argument that width is the most reliable cross-task measure among those considered.
 - Figure 5 clearly visualizes the reported U-shaped trend across three action-decoding architectures under matched train/test granularity, strengthening the generality claim.
 - Figure 6 demonstrates that instruction predictability $P(l|v)$, as probed by cross-entropy of a VLM predictor, improves monotonically with coarseness, supporting the “shallow grounding” hypothesis.
 - Figure 7 shows larger ΔSR under language removal or perturbation for fine and mixed-granularity training than for coarse-only training, indicating reduced language reliance under coarse instructions.
 - Table 5 gives a comprehensive stratified success-rate matrix across training setups and test widths; it also reveals an asymmetric transfer pattern (coarse→fine transfers, fine→coarse struggles).
- The dynamic-instructor setup (Figure 3) is a thoughtful design choice that avoids a single static instruction and better reflects iterative instruction following.
- The paper’s analyses are generally careful, with horizon stratification, modality-attention probing (Table 4), and per-task breakdowns (Figure 9), which add interpretability beyond raw scores.

Summary Of Weaknesses

1. External validity and linguistic realism. The benchmark’s instructions are generated from rule templates grounded in symbolic features (Section 2.3 and Appendix F–G). While this affords control, it limits linguistic diversity and pragmatic phenomena seen in human-written instructions. The U-shaped effect might be partly a property of the template family and feature verbalization style rather than a general phenomenon. There is no experiment with human-authored or paraphrased instructions to test robustness to stylistic variation.
2. Width computability and deployability. The paper concedes (Section 7) that width requires symbolic features and optimal plans, making it primarily an

≡

analytic tool for controlled settings. This sharply limits its applicability in many real-world evaluation regimes where symbolic state is unavailable. The paper would benefit from even a coarse learned estimator of width from text or from learned affordances to test if the findings hold without oracle features.

3. Possible confounds in granularity grouping. Section 4.3 uses per-task quantile binning to form Fine/Medium/Coarse sets. This improves cross-task comparability, but it also means that “coarse” in an easy task can overlap in absolute width with “medium” in a harder task. While the authors explain the motivation, an accompanying analysis that repeats key results with absolute-width bins (e.g., $w=0,1,2,3+$) on the subset of tasks that cover those bins would help rule out binning artifacts. Table 5 reports SR by absolute width but is aggregated across training setups, not isolating the U-shaped claim.
4. Oracle dynamic instruction and supervision leakage risk. The dynamic instructor uses simulator states to pick applicable rules (Figure 3). This could inadvertently simplify credit assignment compared to real-world settings where the agent must infer applicability from partial observations and language. Some ablations where the instructor is noisy, delayed, or only partially state-aware would help bound how much the phenomenon depends on this oracle.
5. Limited coverage of alternative granularity notions. Section 7 mentions other axes like vagueness, loops, or branching, but the paper’s scope is restricted to subgoal decomposition. This is sensible as a first step, yet the conclusions are sometimes phrased broadly. Narrowing claims to subgoal-centric granularity would improve precision, or include a controlled micro-study on one additional axis (e.g., mild vagueness) to check whether the U-shape persists.
6. Language-vision attention analysis interpretability. Table 4’s binomial tests suggest a shift toward visual reliance at the coarse end for Diffusion and a weaker pattern for AR, with Parallel models showing little consistency. However, the paper acknowledges modality entanglement and self-attention confounds. Additional probes, such as gradient-based attribution or causal ablations that mask portions of text vs. image patches at inference, would strengthen the “shallow grounding” interpretation.
7. Success rates are modest even at the peaks. In Figure 5, best matched-train/test SRs hover around 45–49% (Parallel, Coarse/Fine). It would be helpful to discuss practical implications: whether these rates are expected given task horizon (Table 2, Table 10) and whether better absolute SRs would change the qualitative U-shape.
8. Scope of architectures. The study focuses on three action-decoding strategies with a single VLM backbone and LoRA fine-tuning (§H). While this isolates action-decoding effects, it leaves open whether stronger backbones or retrieval-augmented language conditioning could mitigate the medium-

≡

granularity dip. Even a small-scale experiment with an additional backbone or a CoT-augmented variant would help.

9. Absence of human evaluation or user-study evidence. Since the stated motivation involves "diverse human instruction styles," a small human-authored instruction set per task or a paraphrase augmentation experiment could substantially increase confidence that the reported phenomena are not artifacts of rule-to-text templating.
10. Missing closely-related literature on instruction informativeness and language use in embodied RL and agent training. Several relevant works that study the informativeness/diversity of language inputs for agents or examine LLM-based planning for embodied control are not cited (see "Potentially Missing Related Work").

Figure- and table-specific critiques and support:

- Figure 4: The negative correlation between width and SR is consistent across all horizon bins, but effect sizes appear small to moderate (e.g., around -0.26 to -0.20). Reporting confidence intervals on r , not just significance, would help quantify stability.
- Figure 5: The U-shape is visually evident across AR/Parallel/Diffusion under matched granularity. It is persuasive in support of the central claim.
- Figure 6: The monotonic CE loss reduction by granularity is clear, supporting higher $P(I|v)$ for coarser instructions.
- Figure 7: The ΔSR bars make the shallow-grounding point tangible; Pure-C's near-equivalence between "goal-lang only" and "no language" is particularly telling.
- Table 2: The distribution of ranks, lengths, unique tokens, and plan lengths is useful context. However, U_{tok} appears unusually large at lower ranks (e.g., 145 at rank 0 vs. 85 at rank 5); a brief explanation of why lexical diversity decreases with coarsening would be helpful.
- Table 5: The width-stratified matrix is a valuable complement and shows asymmetric transfer. It would be even more compelling if accompanied by a statistical test comparing transfer directions and a breakdown by architecture.

Potentially Missing Related Work

1. Zhang, Yang, Pan, "DANLI: Deliberative Agent for Following Natural Language Instructions," 2022 — Studies neuro-symbolic deliberation for instruction following, relevant to your rule-based representation and planning abstractions. Discuss in Section 2.2 and Related Work; compare conceptual framing around symbolic structures guiding action.
2. Song, Wu, Washington, "LLM-Planner: Few-Shot Grounded Planning for Embodied Agents with Large Language Models," 2022 — Explores using LLMs for planning; relevant to instruction decomposition and granularity. Cite

≡

in Section 3 when motivating metrics and in Section 5 to delineate differences from LLM-derived plan hierarchies.

3. Sumers, Marino, Ahuja, "Distilling Internet-Scale Vision-Language Models into Embodied Agents," 2023 — Connects VLMs and embodied agents; discuss in Section 4.1 as an alternative backbone/teacher signal and in Section 5 to position your action-decoding archetypes.
4. Sarch, Wu, Tarr, "Open-Ended Instructable Embodied Agents with Memory-Augmented LLMs," 2023 — Pertinent to dynamic instructions and dialogue-like guidance; discuss in Section 2.3 and Related Work to contrast dynamic, state-aware instruction with open-ended language interfaces.
5. Wang, Li, Hong, "Bootstrapping Language-Guided Navigation Learning with Self-Refining Data Flywheel," 2024 — Addresses language-guided navigation with evolving data; add in Related Work and discuss whether self-refined language can affect $P(I|v)$ and granularity.
6. Xi, He, Yang, "Teaching Embodied RL Agents: Informativeness and Diversity of Language Use," 2024 — Directly studies types of language inputs and their effects on RL; should be cited in Section 1 and Section 5 when motivating and interpreting mixed-granularity results.
7. Chowdhury et al., "LAGEA: Language Guided Embodied Agents for Robotic Manipulation," 2025 — Uses VLM reflections to guide RL; relevant for the shallow-vs-deep grounding discussion and potential regularizers to improve language reliance (Section 4.4 and Section 5).

These works are closely aligned with instruction informativeness, planning, and embodied language use, and should be integrated into the positioning and discussion.

Comments Suggestions And Typos

Actionable suggestions:

- Add robustness checks with human-authored or paraphrased instructions. Even a small sample per task, or paraphrase augmentation for a subset, would probe whether the U-shaped effect persists beyond the template generator's style.
- Revisit the granularity grouping. Alongside per-task quantiles, report the U-shaped analysis using absolute width bins on the overlapping subset of tasks to confirm the phenomenon is not an artifact of per-task normalization.
- Stress test the dynamic instructor. Add ablations with imperfect instructors: delayed updates, partial observability, or noisy rule selection. This would clarify whether the effect depends on oracle state.
- Strengthen the modality reliance story. Complement Table 4 with causal ablations: mask subsets of instruction tokens or vision patches at inference to quantify sensitivity by granularity. Report effect sizes and confidence intervals.
- Discuss generalization across backbones. Include at least one additional backbone or a simple CoT-enhanced variant, or show that your conclusions are

invariant to modest backbone changes.

≡ Typos and minor issues:

- Page 3/Table 1: The feature list has duplicated “pr” with two meanings (rag and sink distances). Consider renaming one (e.g., ps for sink) to avoid confusion; later text uses ps but Table 1 repeats pr.
- Page 5/Table 3: “ps = X” uses ps, but Table 1 earlier lists pr for sink. Unify notation.
- Page 8/Table 4 caption: “as the it includes self-attention” → “as it includes self-attention”.
- Page 9/References: Arumugam et al. (2025) appears to include a 2017 arXiv entry; correct the year formatting for consistency.
- Page 17/Table 10: Task names have inconsistent underscores and casing (e.g., “washing POTs_and_pans”, “cleaning a car”). Standardize.
- Table 2: “Utok” column label is not explained in-text; later §F.3 uses “Unique Tokens.” Use consistent naming.

Clarifications that may shift my score upward:

- Demonstrate U-shape robustness with absolute-width bins and a paraphrase/human-instruction slice.
- Provide a causal sensitivity analysis to strengthen the shallow-grounding inference.
- Add a small noisy-instructor ablation to reduce concerns about oracle supervision.

Confidence

4: Quite sure after careful checking — I checked the main claims, figures, and tables carefully. Some details, e.g., implementation choices in §H, were not fully re-verified.

Soundness

3.5: Between Acceptable and Strong — The empirical methodology is careful and the analyses are well argued, but reliance on symbolic templates and the lack of robustness checks limit generality.

Excitement

3.5: Between interesting and exciting — The benchmark and the U-shaped finding are memorable and likely to spur follow-up work, especially around “width” as an analysis lens.

Overall Assessment



See how you rank ↑

3.5 Borderline Conference: Solid contribution with notable insights, but missing robustness and breadth to be a clear accept. With added robustness studies (paraphrases/human text, absolute-width replication, noisy instructor), this could be a strong main-track paper.

Best Paper Justification

N/A.

Limitations And Societal Impact

The paper includes a Limitations section acknowledging discrete action spaces, scope of granularity definition, and width computability. Additional points to discuss:

- Linguistic bias due to templated language and symbolic feature verbalization may not reflect natural human instruction styles; this could mislead downstream claims about instruction design in human-robot interaction.
- If the observed “shallow grounding” trend generalizes, overreliance on vision could reduce accessibility in contexts where textual instructions are essential, or encourage neglect of user-provided constraints.
- Sim-to-real concerns: findings may not transfer to real robotic platforms with continuous control and noisy perception. These caveats should be more prominently stated in the main text.

Ethical concerns

None.

Needs Ethics Review

No.

Reproducibility

4: Mostly reproducible — The dataset, rule construction process, and training details are described in reasonable depth, with many specifics in the appendices and tables. Some results may vary due to environment randomness and LoRA fine-tuning choices.

Datasets

4: Useful — If released, the benchmark should be valuable for controlled studies of instruction granularity and grounding, though its templated nature limits broader language research use.

Software



4: Useful — The dynamic instructor, rule bank, and evaluation pipeline will be useful for others studying language grounding under controllable granularity.

Disclaimer: Only the main paper is reviewed; anonymity and formatting are not checked. The desk-rejection assessment evaluates content completeness, correctness, novelty, clarity, and scientific quality in line with common ACL review guidelines.

Reviewed on March 16, 2026  GPT-5

How was this review?

Rate this review to help us improve our AI review quality.

